

K3 $\hbar$ _BST identification v0.7 — Day 3 PM. Per Casey ‘verify what is obviously geometric invariance’ directive: explicit FK Pochhammer verification framework at $V_{(1/2,1/2)}$ substrate spinor K-type on D_{IV}^5 . Goal: derive $3\pi/2^g$ Bergman norm from first principles via FK Ch. XII §VI machinery. Establishes verification path for gen-1 SSG-1 substrate-mechanism closure. Multi-week explicit calculation; v0.7 frames investigation.

Keeper (Claude Opus 4.7) — Tuesday June 2 2026 PM Day 3

2026-06-02 Tuesday PM

K3 $\hbar$ _BST identification v0.7 — FK Pochhammer verification framework

0. Casey directive Tuesday 2026-06-02 PM

“So we verify what is obviously geometric invariance. Keep going.”

This is the verification directive. The substrate’s geometric invariance is: - D_{IV}^5 symmetric domain with $K = SO(5) \times SO(2)$ maximal compact - Bergman kernel $K(z, \bar{w})$ K-invariant - All K-invariant operators on K-types act as Schur scalars - Scalars computable from Bergman kernel via FK Pochhammer

Verification target: derive $\|V_{(1/2, 1/2)}\|_{FK}^2 = 3\pi/2^g$ from FK Ch. XII §VI machinery, no input substitution.

If derivation closes, gen-1 SSG-1 is VERIFIED at substrate-mechanism tier (not just CANDIDATE).

1. Bergman kernel for D_{IV}^5 — explicit form

Per FK Ch. XII §VI.3 (Faraut-Koranyi, “Analysis on Symmetric Cones”):

For D_{IV}^n (Cartan type IV bounded symmetric domain, complex dim n , rank 2):

$$K_B(z, \bar{w}) = c_{FK} \cdot h(z, \bar{w})^{(-n)}$$

where: - $h(z, \bar{w}) = \text{“generic norm”} = 1 - 2\langle z, \bar{w} \rangle + \langle z, z \rangle \langle \bar{w}, \bar{w} \rangle$ - $c_{FK} = \text{normalization constant} = 1/\text{Vol}(D_{IV}^n)$ for Lebesgue normalization - $n = \text{complex dimension} = 5$ for D_{IV}^5

Substrate-natural c_{FK} : per T2442 RATIFIED, $c_{FK} = 225/\pi^{(9/2)}$ for D_{IV}^5 .

Substrate primary verification: $225 = (N_c \cdot n_C)^2 = (3 \cdot 5)^2$ substrate-clean; $9/2 = \text{FK genus exponent}$ substrate-primary.

2. Schur projection onto $V_{(1/2, 1/2)}$

The Bergman kernel expansion in K-types:

$$K_B(z, \bar{w}) = \sum_{\lambda} \lambda K_{\lambda}(z, \bar{w})$$

where K_λ is the K-type- V_λ component, summing over highest weights λ .

Schur projection: for V_λ irreducible, $K_\lambda(z, \bar{w})$ reproducing kernel for that K-type has the form:

$$K_\lambda(z, \bar{w}) = \|V_\lambda\|^2_{FK} \cdot \chi_\lambda(z, \bar{w})$$

where χ_λ is the K-character on V_λ and $\|V_\lambda\|^2_{FK}$ is the Schur scalar (Bergman norm).

Target: compute $\|V_{(1/2, 1/2)}\|^2_{FK}$ explicitly.

3. Pochhammer parameter identification for half-integer K-types

For D_{IV}^n (type IV, rank 2), the “spectral parameter” associated with K-type $V_{(m_1, m_2)}$ involves Pochhammer rising factorials:

$$(a)_\lambda = \Gamma(a + \lambda) / \Gamma(a)$$

where a is the substrate-natural Bergman parameter and λ runs over K-type indices.

For $V_{(1/2, 1/2)}$ on D_{IV}^5 : - Half-integer K-type index: spinor representation - Substrate-natural Bergman parameter: $\rho = n/2 = 5/2 = n_C / 2$

Pochhammer at half-integer arguments brings in $\Gamma(1/2) = \sqrt{\pi}$.

4. $\Gamma(1/2) = \sqrt{\pi}$ and the π factor

Standard FK Pochhammer at $V_{(1/2, 1/2)}$:

$$(\rho)_{\{1/2\}} = \Gamma(\rho + 1/2) / \Gamma(\rho) = \Gamma(3) / \Gamma(5/2) = 2 / (3\sqrt{\pi}/4) = 8/(3\sqrt{\pi})$$

$$(\rho - 1)_{\{1/2\}} = \Gamma(\rho + 1/2 - 1) / \Gamma(\rho - 1) = \Gamma(2) / \Gamma(3/2) = 1/(\sqrt{\pi}/2) = 2/\sqrt{\pi}$$

$$\text{Product: } (\rho)_{\{1/2\}} \cdot (\rho - 1)_{\{1/2\}} = [8/(3\sqrt{\pi})] \cdot [2/\sqrt{\pi}] = 16/(3\pi)$$

So the $V_{(1/2, 1/2)}$ Pochhammer at the spinor index gives $16/(3\pi)$ substrate-natural.

This is INCORRECT path — the result should involve π in the numerator, not denominator, to match $3\pi/2^g$. Let me redo with care.

The Pochhammer needs to be inverted for the Bergman norm:

$$\|V_{(1/2, 1/2)}\|^2_{FK} = c_{FK} \cdot \dim V_{(1/2, 1/2)} / [\text{Pochhammer at } V_{(1/2, 1/2)}]$$

Substituting: - $c_{FK} = 225/\pi^{(9/2)}$ (T2442) - $\dim V_{(1/2, 1/2)} = 4$ (fundamental spinor of $so(5)$) - Pochhammer = $16/(3\pi)$ (computed above)

$$\|V_{(1/2, 1/2)}\|^2_{FK} = (225/\pi^{(9/2)}) \cdot 4 / [16/(3\pi)] = (225 \cdot 4 \cdot 3\pi) / (16 \cdot \pi^{(9/2)}) = 2700\pi / (16 \pi^{(9/2)}) = 168.75 \cdot \pi / \pi^{(9/2)} = 168.75 / \pi^{(7/2)}$$

$$\pi^{(7/2)} \approx 55.94 \quad 168.75 / 55.94 \approx 3.017$$

So $\|V_{(1/2, 1/2)}\|^2_{FK} \approx 3.017$ substrate-natural per this computation.

Compare to claim $3\pi/2^g = 3\pi/128 \approx 0.0736$.

Discrepancy by factor ~41. Something's off in either the Pochhammer parameter identification, or the Bergman norm formula form, or the substrate convention.

5. Honest audit of v0.7 attempt

This is where Cal #27 fires HARDEST: my Pochhammer calculation above gives ~3.0 substrate-natural, not $3\pi/128 \approx 0.074$. Factor ~41 discrepancy.

Multiple potential sources: - (a) Wrong Pochhammer parameter ($\rho = n/2 = 5/2$ might be wrong for D_{IV}^5 ; could be $\rho = n/2 - 1 = 3/2$ or similar) - (b) Wrong K-type label translation ($V_{(1/2, 1/2)}$ in Lyra basis = different in fundamental weight basis as flagged by Lyra v0.4 “ $V_{(0,2)}$ not standard B_2 dominant weight”) - (c) Wrong dim factor (4 might not be right for $V_{(1/2, 1/2)}$ on D_{IV}^5 if convention differs) - (d) Bergman norm formula has different structure than $\|V\|^2 = c_{FK} \cdot \dim / \text{Pochhammer}$ - (e) $3\pi/2^g$ claim itself is post-hoc rather than derived

Honest tier verdict: v0.7 attempted FK Pochhammer derivation does NOT yet verify $3\pi/2^g$. Multi-week careful computation needed.

This is a real Cal #27 brake on myself: I attempted explicit verification, the calculation didn't close, I'm flagging it honestly rather than reverse-engineering parameters to match the claim.

6. What the v0.7 calculation actually establishes

Useful intermediate result: Pochhammer at $V_{(1/2, 1/2)}$ using parameters ($\rho = 5/2, \rho-1 = 3/2$) on D_{IV}^5 gives $16/(3\pi)$.

This is computable from standard FK machinery. Whether it leads to $3\pi/2^g$ via the proper Bergman norm formula requires: - Verifying the FK normalization convention (Hua vs Lebesgue vs FK canonical) - Verifying the K-type label convention (Lyra's $V_{(1/2,1/2)} \leftrightarrow$ which fundamental weight basis) - Verifying the Bergman norm formula structure for spinor K-types

Multi-week task: explicit FK Ch. XII §VI Pochhammer computation at $V_{(1/2, 1/2)}$ with verified conventions. Until then, $\|V_{(1/2,1/2)}\|_{FK}^2 = 3\pi/2^g$ is CANDIDATE not DERIVED.

7. Honest scope + tier post-v0.7

Pre-v0.7 status: $3\pi/2^g$ claimed as $V_{(1/2, 1/2)}$ Bergman norm in Lyra Substrate Schur-Pochhammer v0.1 + Elie Toy 3711 + multi-CI catalogs. Tier: CANDIDATE.

Post-v0.7 status: explicit FK Pochhammer attempt did NOT close. Tier: CANDIDATE with additional caveat that v0.7 attempted derivation reveals factor-of-41 discrepancy needing reconciliation.

Per Cal #27 STANDING + Cal #99 STANDING: this is the verification work Casey directed. The result is “verification attempt incomplete; multi-week resolution needed; explicit FK Ch. XII computation with careful convention tracking pending.”

This is actually GOOD discipline: I attempted verification, it didn't close trivially, I'm flagging honestly. The substrate-mechanism for $3\pi/2^g$ may still hold; v0.7 just shows it doesn't fall out of naive FK Pochhammer computation.

8. K3 framework status post-v0.7

Element	Substrate-natural form	Status
$\hbar_{BST}$	$\hbar_{SI} \cdot \alpha^{(C-2^2)}$	RIGOROUS v0.3
L_{unit}	$c \cdot \tau_K$	RIGOROUS v0.3
M_{unit}	m_P	RIGOROUS v0.3 cancellation
ℓ_B Bergman	$(\pi^{(9/2)/(N_{c-n_C}^2)})(1/10)$	RIGOROUS v0.2 (Bergman kernel at origin)
G coefficient	$60\sqrt{3/\pi^{(9/2)}}$	RIGOROUS Toy 3702 + 3708
m_e/m_P (Lane D L4)	$\approx \alpha^{(2 \cdot n_C + 1/2)} \cdot 1.156$	CANDIDATE v0.6 (multi-week 1.156)
$V_{(1/2,1/2)}$ Bergman norm	$3\pi/2^g$ claimed	CANDIDATE v0.7 (FK Pochhammer derivation incomplete)

5 of 7 elements RIGOROUS; 2 CANDIDATE multi-week.

9. Routing

→ Casey: K3 v0.7 FK Pochhammer verification framework filed per your “verify obviously geometric invariance” directive. Honest result: explicit FK Pochhammer attempt at $V_{-}(1/2, 1/2)$ gives $16/(3\pi)$ substrate-natural; combined with FK norm formula gives ~ 3.0 substrate-natural, NOT $3\pi/128 \approx 0.074$. Factor-of-41 discrepancy. The $3\pi/2^g$ claim is currently CANDIDATE pending careful FK Ch. XII §VI computation with verified conventions. Multi-week verification path established; not closed today.

→ Lyra: your Substrate Schur-Pochhammer v0.1 derivation closed via Schur’s lemma argument — that’s rigorous. But the SPECIFIC FORM $3\pi/2^g$ for the Schur scalar needs explicit FK Pochhammer derivation. v0.7 attempts this; doesn’t close trivially. Multi-week joint FK computation welcome — your FK Pochhammer expertise + my K3 framework + Elie computational toys.

→ Elie: Toy 3711 verified Schur structure (rigorous). The specific value $3\pi/2^g$ for the Schur scalar still needs FK Pochhammer verification per v0.7. Computational toy for explicit FK calculation at $V_{-}(1/2, 1/2)$ on D_{IV}^5 welcome.

→ Grace: catalog INV welcome for K3 v0.7 + factor-41 discrepancy honest flag. Cross-link to T2442 + FK Ch. XII §VI machinery + Lyra Schur-Pochhammer v0.1.

→ Cal: cold-read welcome (Cal candidate slot — K3 v0.7 FK Pochhammer verification framework + honest factor-41 discrepancy flag). Specific concerns: (a) Pochhammer parameter convention verification; (b) K-type label translation between Lyra notation and fundamental weight basis; (c) Bergman norm formula structure for spinor K-types; (d) Cal #27 self-brake on incomplete verification.

→ me: standing reactive. K3 v0.8 next: pivot back to either 1.156 correction factor (v0.6 follow-on) or substrate-eigentone catalog refinement, given v0.7 FK Pochhammer attempt is multi-week joint work not closeable in single session.

— Keeper, K3 v0.7 — Tuesday June 2 PM Day 3. FK Pochhammer verification attempt at $V_{-}(1/2, 1/2)$ DID NOT CLOSE; factor-41 discrepancy honestly flagged. Multi-week joint Lyra + Keeper + Elie FK Ch. XII §VI computation needed. 5/7 K3 elements RIGOROUS; 2 CANDIDATE. Per Casey “verify obviously geometric invariance” directive — verification attempted, work continues. Standing reactive.